

Activity 13 · Claude Code Skills and MCP Audit

C386 · v1.0 · Synthetic training data

Goal

A reproducible research workspace stores evidence, code and review rules. A local read-only MCP connector or CSV fallback supplies synthetic channel metrics; a custom Skill defines research checks without granting tools or overriding permissions.

Before you start

Use a copied activity folder, Claude Cowork or Claude Code as specified, and a spreadsheet or Python3 for independent checks. No production credentials are needed. Product access varies; CSV fallback remains valid.

Mock data and assets

mock-data/tool-results.csv

All named customers, competitors, campaigns and values are fictional. Assets include brand, source, measurement and decision contracts.

Mechanism

User question -> reviewed Skill -> approved read-only MCP tool -> schema validation -> deterministic calculation -> source-linked report -> reviewer. CSV fallback tests the same data contract.

The Cowork ebooks and Coupler custom-workflow examples inform reusable instruction packaging; current Anthropic Code documentation governs Skill paths and MCP configuration. Windsor's analytics context motivates currency, time-zone and grain checks before aggregation.

Worked calculation

Lead rates: paid 5%, organic 4%, email 8%, direct 1%; total $96/2400=4\%$. A Skill's instructions do not create connector access.

Step-by-step

1. Read assets/research-context.md and assets/market-evidence-audit/SKILL.md; inspect the read-only MCP server source.
2. Copy the activity folder to a dedicated working directory; open a terminal there and run `python3 validate.py` for actual CSV metric checks.
3. Run `python3 test_mcp.py`. Verify actual local initialize, tools/list and tools/call responses; no live account is connected.
4. For Claude Code, use the current official installation and launch claude inside the copy. Place the reviewed Skill under `.claude/skills/market-evidence-audit/SKILL.md`.
5. Register the local server only after reviewing its single read-only tool: `claude mcp add --transport stdio --scope project synthetic-market-metrics -- python3 "$PWD/mcp_metrics_server.py"`. Review the project configuration before approving.
6. Within Claude Code, inspect /mcp and allow only `read_synthetic_metrics`. Submit Prompt1 and compare returned source IDs M01–M04 to the CSV.

7. For Cowork custom Skills,zip assets/market-evidence-audit with SKILL.md at the root;ensure Code execution and file creation is enabled,then open Customize > Skills > + > Create skill > Upload a skill and upload the ZIP. Enable it and verify the task actually used it;copying .claude files does not install a Cowork Skill.

8. Review generated code,run its calculation,and verify5%,4%,8%,1%,overall 4%. Save output,tool provenance and human review;state local protocol test versus actual Claude connector invocation separately.

Prompt1

Build a reproducible synthetic research analysis. Read the supplied Skill and tool-results.csv. Preserve source IDs, validate schema and calculate rates. If MCP is unavailable, use CSV fallback and record it. Do not install or connect to a live account without authorisation.

Prompt2

Challenge the previous result. Find denominator errors,unsupported claims,bias,privacy risks and unexecuted tool boundaries. Cite exact row IDs or fields. Revise the report without inventing data.

Acceptance evidence

Four source IDs retained; rates 5%,4%,8%,1%,overall 4%; local MCP initialize/list/call test passes; Skill front matter valid; Claude invocation separately evidenced;no live account claimed.

Failure and recovery

The agent treats the Skill as permission to access a production account. Instruction reuse and tool authorisation are separate controls.

Inspect tool scope and data destinations; use read-only synthetic connector, no secrets in repository. Require approval for new network endpoints and use local fallback if account access is unavailable.

Submit

SKILL.md,analysis.py and tool-audit.md

Save outputs in this working copy. Keep an execution note stating which product actions actually ran and which used fallback.